Microplastic Size in Rivers and Ponds, and the University of Nottingham Malaysia Contribution

Literature screen of the filtered Bibliometrix dataset

Group Project – Case Studies (Combined)

November 2, 2026

Abstract

This report answers two questions from the uploaded Bibliometrix dataset (`Bibliometrix_Filtered.RData`): (1) what microplastic size is most commonly reported in rivers and ponds, with a focus on the Semenyih River; and (2) which papers are affiliated with the University of Nottingham Malaysia (UNM), what methods they used, and what they found. Every count below comes from a documented filter applied to the records in the file, and the filter code and output tables are included in the package. In the screened river/lake/pond abstracts that state a dominant size class (n = 616), about 87.3% report a class below 1 mm. UNM contributes nine records, of which five are primary microplastic studies in the Langat–Semenyih catchment. **Update:** the full texts of two papers (Chen et al. 2021 on the Langat River and the 2026 review of Malaysian rivers) are now used, which supplies UNM instruments (Leica EZ4 stereomicroscope, PerkinElmer Frontier FTIR), a 53 μm sieve cut-off and a reported visual-identification accuracy of 96%. The other UNM papers remain abstract-only, and no source gives a size histogram for the Semenyih River itself.

1 Data provenance and the 10,833 vs 10,897 discrepancy

The uploaded file contains one data frame (`M`) with 10,897 records and 68 fields. You mentioned 10,833 papers; I could not reproduce that figure from this file (e.g. de-duplicating on DOI gives 10,890 and removing records without an abstract gives 10,853). All counts here therefore start from 10,897 records, and the funnel shows every step so the base can be adjusted if a different starting set is intended. Of the records, 8,800 originate from Web of Science and 2,097 from Scopus. Two full-text PDFs were added afterwards: Chen et al. 2021 [6] and Bouida et al. 2026 [5]. Where a statement comes from them it is marked "full text"; everything else still comes from abstracts.

2 How the papers were filtered

2.1 Funnel for Question 1 (size in rivers and ponds)

Filters were applied to the concatenated title, abstract and author keywords (upper-cased, micro sign normalised). Exact patterns are in Appendix A and in `scripts/filter_pipeline.py`.

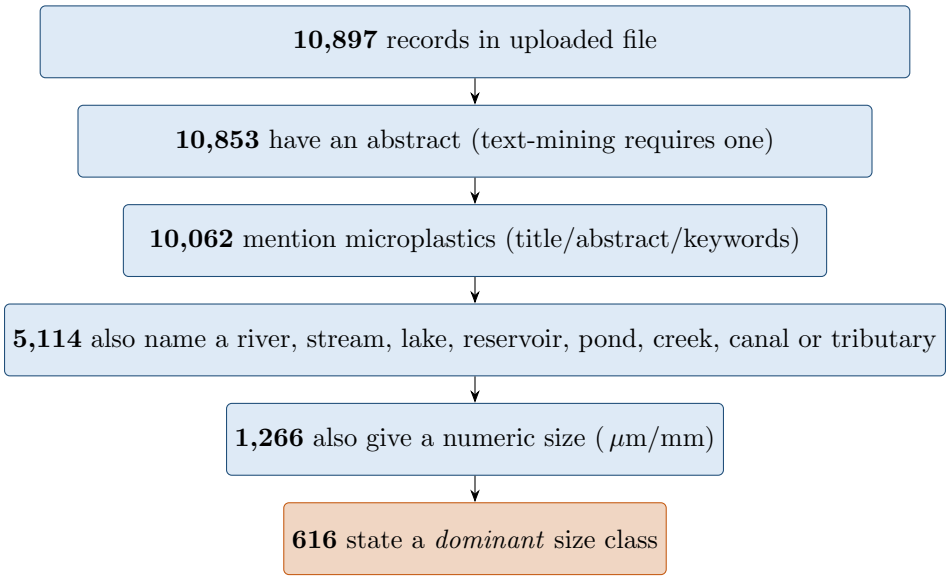


Figure 1: Filter funnel for Question 1. A broader "freshwater/wetland" wording would keep 7,236 records at the water-body step; it was not used because it admits non-river/pond systems.

How the last step works: for each abstract that passed the numeric-size step, sentences were kept only if they contained a dominance word (dominant, majority, most, predominant, mainly, prevalent, etc.) and a size expression. Sentences that merely define microplastics (<5 mm, 1 μm–5 mm) or give a study's search range were removed. The first remaining expression was converted to millimetres and assigned a class by its *upper bound*: ≤0.1 mm, 0.1–0.5 mm, 0.5–1 mm (this class also holds generic "<1 mm" statements) and >1 mm up to 5 mm.

2.2 Filter for Question 2 (University of Nottingham Malaysia)

Six affiliation fields (`C1`, `affiliations`, `AU_UN`, `RP`, `AU1_UN`, `organization`) were searched for "Nottingham" within 80–200 characters of "Malaysia", "Semenyih", "Broga" or "Selangor", plus the abbreviation UNMC. This returned **9 records**. For comparison, 33 records mention Nottingham in `C1`, but most belong to the UK campus, Nottingham Ningbo China, Nottingham Trent University or the British Geological Survey, and were excluded. A cross-check on author names (Chen HL, Selvam SB, Gibbins C, Ting KN, Ng YS, Show PL) found two further records not in the set: a Ningbo-campus review and an unrelated schlieren-imaging paper; neither lists a Malaysian Nottingham address.

3 Question 1 – Literature review: common microplastic size in rivers and ponds

3.1 What the screen shows

Among the 616 river/lake/pond abstracts that state a dominant class, sub-millimetre particles dominate (Figure 2). Classes by share: <100 μm 16.2%, 100–500 μm 37.5%, 0.5–1 mm 33.6%, and 1–5 mm 12.7% (counts 100, 231, 207, 78). The pattern is similar in rivers (n = 472; 86.9% sub-millimetre), lakes and reservoirs (n = 212; 89.2%) and ponds (n = 25; 96.0%, small sample so indicative only). The abstracts also report that fibres are usually the dominant shape in rivers.

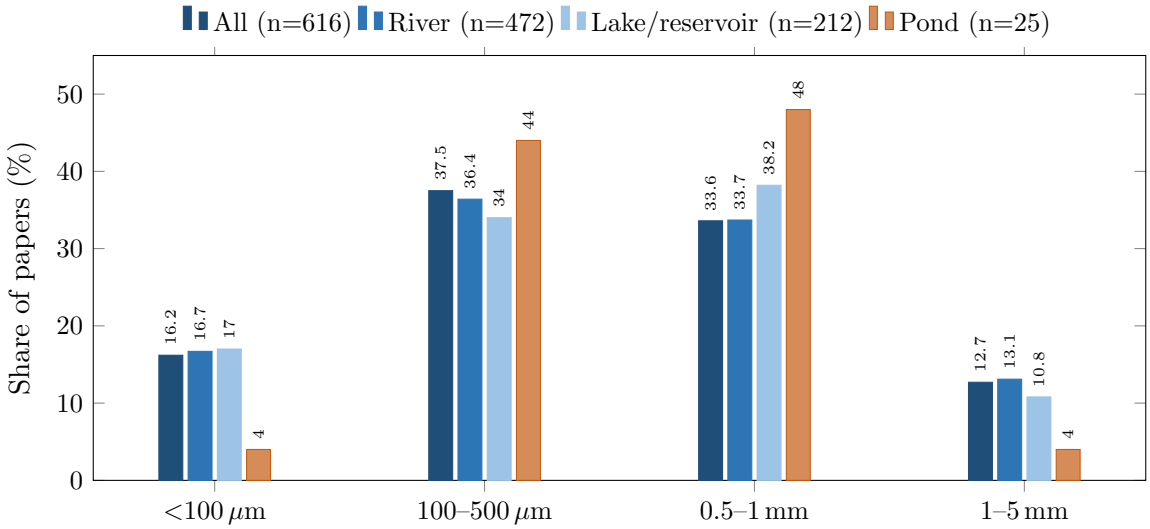


Figure 2: Dominant size class reported in abstracts (river/lake/pond microplastic papers). A paper can appear in more than one water-body group.

3.2 Why the answer depends on the method

Reported size is partly a function of what the method can see. Within the UNM group's own fish/invertebrate study, visual inspection covered 0.1–5 mm while a new automated method covered 0.004–0.1 mm and returned substantially higher counts, so visual work under-estimates small particles [16]. A Perak River study using fluorescence and FTIR microscopy found most particles at 1–10 μm [12], and a review of Malaysian rivers notes that detection limits fell to 1–10 μm as the field moved from visual microscopy to μFTIR and Py-GC/MS [5]. The headline "most are below 1 mm" is therefore robust, but the split between 100–500 μm and 0.5–1 mm depends on mesh and instrument.

3.3 Malaysian and pond examples from the screen

Study	Water body	Dominant size stated in abstract
Zaki et al. 2024 [20]	Selangor River, water	600–1000 μm (26.3%); fibres 64.3%; stereomicroscope + FTIR
Zaki et al. 2023 [19]	Selangor River, sediment	<300 μm (more than half of particles)
Ali et al. 2025 [1]	Langat River	10–1000 μm; mostly fragments
Anuar et al. 2023 [3]	Langat and Kelantan	Sizes classified; values not in abstract; μFTIR, Py-GC/MS, SEM-EDS
Sheriff and Adams 2026 [18]	Review, Malaysia	Langat particles often below 1 mm
Hassan et al. 2025 [12]	Perak River intake	Mostly 1–10 μm (fluorescence/FTIR microscopy)
Cherating River 2020 [10]	River + mangrove	Fragments 0.5–1.0 mm
Klang estuary 2021 [13]	Estuary water	300–1000 μm
Baram estuary 2021 [4]	Estuary	0.3–1 mm
Urban ponds, Raipur 2024 [17]	20 tropical ponds, India	500–1000 μm class prevailing

Table 1: Selected regional and pond studies. Values are taken from abstracts only.

4 Question 1.1 – Focus diagram: microplastic size in the Semenyih River

What the sources can and cannot say. The Semenyih River is one of the two similar-sized upper sub-catchments that merge to form the Langat; Chen et al. 2021 placed sites 1–3 on the Semenyih River (site 1 about 1 km below the Semenyih dam, site 2 at the upstream edge of Semenyih town, site 3 just downstream) and sites 4–8 along the Langat mainstem [6] (full text). Sites 1–3 had significantly lower concentrations than the downstream sites, and microplastics were still found at site 1 near the dam. The Semenyih-specific study of Chen et al. 2023 sampled eight sites on the upper Semenyih River, five replicates of water and bed on three dates (160 samples), with 3.12 ± 2.49 particles/L in water, $6027 \pm 16{,}586$ particles/m^2 on the bed and fibres dominant [9]. Neither source reports a size histogram. What they do fix is the *counted range*: both pass samples through a 53 μm sieve and count by stereomicroscope (8–35×), so particles smaller than about 53 μm are outside the data, and the review lists stereomicroscopy as typically reliable above 300 μm [5]. The diagram therefore shows this counted range for the Semenyih work next to the Langat and Selangor size results and the UNM method windows.

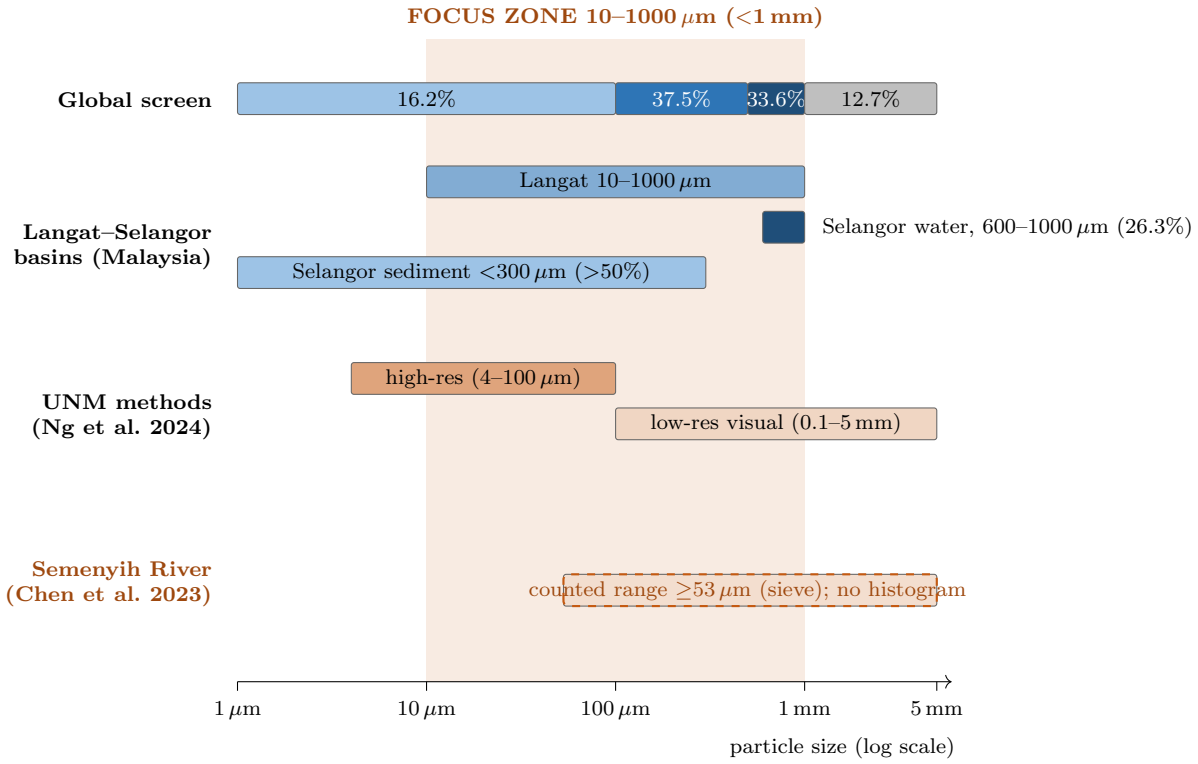


Figure 3: Focus diagram. Top row: share of the 616 screened papers by dominant class. Middle: dominant classes reported for the neighbouring Langat and Selangor basins [1, 20, 19]. Orange: size windows of the UNM group's low- and high-resolution methods [16]. Bottom: counted range of the UNM Langat/Semenyih work, limited below by a 53 μm sieve [6, 9]; neither gives a size histogram. The shaded zone marks where the regional evidence concentrates.

Reading the diagram. All regional evidence concentrates below 1 mm, with water samples centred around 0.3–1 mm and sediment skewed finer (<300 μm). Because the UNM group's own work shows that a visual method misses the finest particles [16], the Semenyih size distribution should be read as a lower bound on the fine fraction. Table 2 shows what the two full-text sources supply and what is still missing.

Item to extract (Chen et al. 2023 [9])	Why it is needed	Value
Size-class table, water (% per class)	Would replace the dashed range with real bars	Not reported in either source
Size-class table, bed sediment	Water vs bed comparison (fragments rose on the bed)	Not reported in either source
Smallest size counted	Lower edge of the bar	53 μm sieve; 1.2 μm GF/C filter (full text)
Identification instrument	Detection method and accuracy	Stereomicroscope, hot-needle test, FTIR check (Section 5.1)

Table 2: Status of the items needed to complete the Semenyih size bars (Chen et al. 2021 and 2023 [6, 9]).

5 Question 2 – Papers affiliated with University of Nottingham Malaysia

Ref.	Year	Title	Journal	Cites	Role in this review
[6]	2021	Spatio-temporal variation of microplastic along a rural to urban transition in a tropical river*	Environmental Pollution	69	Core – river monitoring
[7]	2021	Microplastic pollution in freshwater systems in Southeast Asia: contamination levels, sources, and ecological impacts	Environmental Science and Pollution Research	38	Review (SE Asia)
[8]	2022	Relative contributions of different local sources to riverborne microplastic in a mixed landuse area within a tropical catchment	Environmental Research	18	Core – source apportionment
[9]	2023	Microplastic concentrations in river water and bed sediments in a tropical river: implications for water quality monitoring	Environmental Monitoring and Assessment	15	Core – river monitoring
[14]	2022	U-net skip-connection architectures for the automated counting of microplastics	Neural Computing & Applications	28	Core – image analysis
[16]	2024	Micro- and nano-plastic loads in fish and macroinvertebrates in a tropical river	Freshwater Biology	4	Core – biota / method comparison
[11]	2023	Weathering effect triggers the sorption enhancement of microplastics against oxybenzone	Environmental Technology & Innovation	39	Peripheral lab study (Taiwan-led)
[15]	2021	Source, distribution and emerging threat of micro- and nanoplastics to marine organism and human health: socio-economic impact and management strategies	Environmental Research	117	Peripheral review (Show PL)
[2]	2026	Interactions between phytoplankton species and micro/nano-plastics and heavy metal contamination	Biological Reviews	4	Peripheral review (Show PL)

Table 3: The nine records returned by the UNM affiliation filter (citation counts are the `TC` field in the dataset).

Five records are primary microplastic studies (river monitoring, source apportionment, biota and image analysis); one is the group's Southeast Asia review. The three others carry a UNM address through the Department of Chemical and Environmental Engineering (Show PL) and cover marine/health reviews, phytoplankton and a Taiwan-led aging and sorption experiment, so they are less relevant to river detection.

5.1 Methodology for detecting microplastics, UNM equipment and accuracy

Scope of evidence. The Langat paper (Chen et al. 2021) is covered from its full text. For the other papers, only abstracts (and, for Chen 2022 and 2023, the review's method table) are available, so "not stated" means the files I have cannot answer it.

Paper	Sampling and detection stated	Instrument named?	Accuracy / performance figure in abstract
Chen 2021 (Langat; *full text*) [6]	8 mainstem sites, Nov 2019–Oct 2020; 5×2 L depth-integrated samples per site; field sieve 53 μm; H_2O_2 + Fe(II) digestion at 70°C; GF/C 1.2 μm filters; visual count at 8–35×	Leica EZ4 stereomicroscope; PerkinElmer Frontier FTIR (transmission, 4 scans, 4000–400 cm^{-1}) on 85 picked particles	**96%** visual-identification accuracy (FTIR check, n = 85); blanks processed, no contamination found
Chen 2023 (Semenyih) [9]	160 water and bed samples; per review table: 2 L bulk water, bed resuspension in a rigid cylinder, 53 μm prefilter, NOAA-type peroxide oxidation at 60°C, $ZnCl_2$ density separation for bed [5]	Stereomicroscopy with hot-needle test [5]	Blank subtraction (water column vs cylinder) [5]; within-site variability often exceeded between-site
Chen 2022 (sources) [8]	Culvert/drain water, atmospheric deposition, five WWTPs; polymers identified (styrene-butadiene rubber, nylon)	Spectroscopic identification implied, not named	WWTP removal efficiency 30.95±5.51% to 69.94±22.17% (treatment performance, not detection accuracy)
Ng 2024 (biota) [16]	Fish and macroinvertebrates; low-resolution visual inspection (0.1–5 mm) and a novel automated method (0.004–0.1 mm)	"Novel automated method" (type not named)	High-resolution method gave substantially higher counts than visual; no % given
Lee 2022 (U-Net) [14]	Microscope images; U-Net segmentation plus post-processing to count particles; skip-connection variants	Microscope imaging (model not named)	"Human-level performance" counting in cluttered images; no numeric score in abstract
Hanun 2023 (aging) [11]	PET microplastics aged in air, seawater and freshwater; SEM, FTIR, BET	SEM, FTIR, BET (instruments at the Taiwan-led lab, not UNM)	Not a detection study; sorption 6.62–9.10 mg/g aged vs 6.44 mg/g virgin

Table 4: Methods and performance as stated in the abstracts of UNM microplastic papers.

UNM laboratory equipment and workflow (Chen et al. 2021, full text).

Stage	What was used
Sampling	2 L bottles lowered through the water column at five points across the channel (10 L per site per date); a pilot at site 8 showed lateral variation, so full-width sampling was adopted
Pre-sieving	Stainless-steel sieve, 53 μm mesh, in the field
Digestion	20 mL of 30% H_2O_2 plus 20 mL of 0.05 M Fe(II) catalyst, heated to 70°C for 30 min (NOAA method of Masura et al.); re-sieved at 53 μm
Filtration and drying	Whatman GF/C glass-fibre filters (1.2 μm), vacuum pump, oven-dried at 100°C for 24 h
Counting	Stereomicroscope (Leica EZ4), 8× to 35×, particles per litre
Polymer check	PerkinElmer Frontier FTIR, transmission mode, 4 scans, 4000–400 cm^{-1}, matched to reference spectra
Contamination control	Cotton clothing and lab coats, nitrile gloves, covered samples, rinsed apparatus; field and lab blanks gave no contamination
Suspended sediment	EPA 160.2 gravimetric method, GF/C filters dried at 100°C

Table 5: Equipment and procedure reported for the UNM Langat study [6].

Accuracy. The only detection-accuracy figure found is the Langat study's FTIR check: 85 randomly picked particles were scanned and the visual identification was reported as 96% accurate; the polymers were PET 43%, HDPE 21%, LDPE 13%, PP 8%, EVA 8%, PVDC 5% and PS 4%, and five particles were non-synthetic (cotton, viscose, silica) [6]. This test validates identification, not size or recovery: no spiked-particle recovery rate or detection limit is given. The review adds that stereomicroscopy has low polymer-identification reliability unless confirmed by a hot-needle test, and that its usual size limit is about 300 μm [5]; the group's later automated method reaches down to 4 μm [16], and a deep-learning counter reaches human-level counts [14]. Instrument details for the Ng, Lee and Chen 2022/2023 papers still need their own full texts.

5.2 Summary of UNM findings

Abundance. Langat mainstem mean 4.39 ± 5.11 particles/L over 508 samples, rising more than threefold from the rural upper reach to the urban lower reach (sites 7 and 8 highest); the two industrial tributaries averaged 45.86 ± 24.76 and reached 90 particles/L [6] (full text). The river was estimated to convey about 5 billion particles/day half of the time and about 30 billion/day for 10% of the time. The Semenyih River carried 3.12 ± 2.49 particles/L and $6027 \pm 16{,}586$ particles/m^2 on the bed [9]. The review cautions that the very high Langat values of other groups (up to 2012 items/L) reflect finer detection (0.2 μm filter, μFTIR) rather than a pure environmental difference [5].
Type and origin. In the Langat, fibres were 96%, fragments 3%, films 1% and beads under 1%, mostly secondary; domestic textile washing and aerial fallout are the likely sources in the near-forest upper reach, and clothing-factory drains in the tributaries [6]. In the Semenyih, fibres dominated water and bed with more fragments on the bed [9], and the review reports PE and polyester there and a "sink danger" risk category [5]. Animals held a more even mix (about 35% fibres, 43% fragments, 22% film) [16].
Drivers. Concentrations averaged 1.53 ± 0.87 particles/L at low flow and 17.83 ± 6.73 particles/L at elevated discharge, tripled during a June 2020 flood, and tracked suspended sediment (quantile regression pseudo-R^2 0.1 for the median and 0.65 for the 90th quantile), so turbidity sensors could estimate loads [6]. Road culverts delivered 42.20 ± 35.29 particles/L, up to seven times residential (8.53 ± 9.91) and industrial (5.67 ± 4.88) drains; atmospheric deposition was 76.07 ± 32.85 particles/m^2/day; WWTP outflows were similar in concentration to uncontrolled drains [8].
Biota. 95% of fish and 44% of macroinvertebrates held larger microplastics, with loads linked to feeding group and body size [16].
Monitoring advice. Water-only monitoring may misrank bed contamination, and single bed samples are inadequate [9]; high-resolution processing is needed for small organisms [16].
Regional context. The group's review concludes that data in Southeast Asian fresh waters are limited, with Malaysia a focus, and lists transport, fate and effects as gaps [7]. Show PL's reviews cover health, socio-economic effects and phytoplankton interactions [15, 2].

6 Limitations

- Text-mining of abstracts (Question 1): papers that report size only in the full text are missed, and a few off-target hits (for example soil or snow studies that mention a river) remain in the 616.
- Classes are assigned by upper bound, so a 300–1000 μm statement falls in the 0.5–1 mm class.
- The Semenyih size histogram is still unknown: the full text of Chen et al. 2023 itself is needed. Methods for that paper and for Chen et al. 2022 are second-hand, from the review's summary table [5].
- The Chen et al. 2021 accuracy figure is quoted as stated (96%); 5 of the 85 tested particles were non-synthetic, which is arithmetically 94.1% correct. The review's Table 3 lists granules as the dominant Semenyih shape while its text and the Semenyih paper say fibres; the review's text also gives the Langat range as 1811–4157 items/L where the source abstract says 181–4157.
- Starting count is 10,897, not 10,833 (Section 1).

A Filter rules

Step	Pattern (case-insensitive, on title + abstract + keywords)
Microplastic term	`MICRO-?PLASTIC | MICROFIBER | MICROFIBRE | PLASTIC PARTICLE | PLASTIC DEBRIS | MICRO- AND NANO`
Water body	`RIVER | POND | LAKE | STREAM | RESERVOIR | CREEK | CANAL | TRIBUTAR`
Numeric size	number + unit (μm, mm, nm), as a range, a "<" form or a single value
Dominance word	`DOMINAT | MAJORITY | MOST | PREDOMIN | MAINLY | PREVAIL | PREVALEN | COMMON | ABUNDANT | ACCOUNT`
Definition exclusion	sentences defining microplastics or giving a search range (e.g. "<5 mm in size")
UNM affiliation	`NOTTINGHAM` within 80–200 characters of `MALAYSIA`, `SEMENYIH`, `BROGA` or `SELANGOR`; or `UNMC`

B Package contents

- `report.tex` / `report.pdf` – this document.
- `scripts/filter_pipeline.py` – reproduces every count and table from the RData file.
- `data/q1_dominant_size_papers.csv` – the 616 papers with the matched sentence, size expression and class.
- `data/q1_river_pond_lake_microplastic_papers.csv` – the 5,114 papers after the water-body step.
- `data/q2_unm_papers.csv` – the nine UNM records with affiliations and abstracts.
- `data/funnel_q1.json` – funnel counts.

References

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